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Surgery Followed by Cornea-Sparing Treatment Approach with Proton Therapy Reduces Orbital Exenteration for Cancer Patients with Periorbital Malignancies



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Purpose/Objective(s): Periorbital malignancies are uncommon and technically challenging to treat given the proximity of several critical structures of the head and neck. Proton therapy offers a dosimetric advantage over photon-based radiation therapy (RT) by reducing dose exposure to tissues beyond the target range. Herein, we present the management, toxicity data, and outcomes after proton therapy for patients with periorbital head and neck cancers.

Materials/Methods: Patients with primary, non-metastatic periorbital cancers treated with proton RT at our institution from 2008 to 2018 were included in this study. Descriptive statistics were used to evaluate patient and tumor characteristics as well as acute and chronic toxicities after proton RT. Two- and 5-year locoregional control (LRC), disease-free survival (DFS), and overall survival (OS) were estimated using the Kaplan-Meier method.

Results: Forty-four patients met the inclusion criteria and had a median age of 56 (range, 19-78) years with a median follow up time of 4.5 years. Most patients were white (n=32, 73%), male (n=30, 68%), and had clinical T2 or T3 cancers (n=27, 61%) of the lacrimal sac or gland (n=34, 77%). An orbit-sparing surgical approach was used for 39 patients (89%), and all but 2 patients were treated with surgery followed by (chemo-)radiation. There were 63% grade 1, 30% grade 2, and 6% grade 3 acute toxicities reported, most of which were radiation-induced dermatitis (grade 3: 80% of cases). In patients that had their orbit spared, xerophthalmia was the most common grade 1 ocular toxicity (n=11, 28% of patients) followed by diplopia and epiphora (15% each). Rates of chronic grade 3-4 toxicities were low at 8% (grade 4: 1 corneal ulcer and 1 retinopathy). Two- and 5-year LRC, DFS, and OS were 81% and 81%, 73% and 61%, and 77% and 63%, respectively.

Conclusion: Multimodal therapy that includes proton radiation therapy results in excellent locoregional control and survival with preservation of vision and cosmesis for patients with periorbital malignancies.

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Ruthenium-106 Plaque Brachytherapy for Ocular Tumors: A Review of 327 Cases



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Purpose/Objective(s): To study the long-term outcomes of Ruthenium-106 (Ru-106) plaque brachytherapy for ocular tumors in the Indian population.

Materials/Methods: Retrospective review of case records of 327 patients who were treated with plaque brachytherapy between January 2012 and December 2017.

Results: 327 patients with uveal melanoma (n=112), choroidal hemangioma (n=91), retinoblastoma (n=44), ocular surface squamous cell neoplasia (OSSN n=56), choroidal metastasis (n=10), conjunctival melanoma (n=4), retinal capillary hemangioma (n=3) and others (n=8) were treated with Ru-106 plaque brachytherapy (notch= 217, round= 110). The mean dose of radiation was 9863±801.6cGy for uveal melanoma, 3856±604.9cGy for choroidal hemangioma, 4747±667.3cGy for retinoblastoma and 5275±545.2cGy for OSSN. Tumor regression was noted in 85%, 97%, 87%, and 56% cases of uveal melanoma, choroidal hemangioma, OSSN, and retinoblastoma respectively. Eye salvage was achieved in 87%, 60%, 91%, and 100% cases of uveal melanoma, retinoblastoma, OSSN, and choroidal metastasis respectively. Visual acuity improved or remained stable in 91%, 94%, 62%, and 53% cases of choroidal hemangioma, retinoblastoma, OSSN, and uveal melanoma respectively. Life salvage was 99% in cases with uveal melanoma and 100% for OSSN. Radiation retinopathy developed in 6% cases with uveal melanoma and 6.5% with choroidal hemangioma. Cataract was seen as a complication in 6% patients with uveal melanoma. 14% of patients with retinoblastoma had vitreous hemorrhage post radiation.

Conclusion: Ru-106 plaque brachytherapy is a safe and effective treatment modality for a multitude of ocular tumors with good outcomes in terms of life, eye, and vision salvage.

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Characterization of the Oropharyngeal Microbiome during Radiotherapy in HPV-Associated Oropharynx Cancer



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Purpose/Objective(s): The purpose of this study was to describe the baseline and serial oropharyngeal microbiome in HPV-associated OPC over the course of radiotherapy (RT).

Materials/Methods: Patients newly diagnosed HPV-associated OPC treated with definitive radiotherapy +/- concurrent chemotherapy were enrolled in this prospective study. Using 16S rRNA gene sequencing, dynamic changes in mucosal microbiome during RT were investigated. Oropharyngeal samples were obtained before RT and at week 1, 3 and 5 of RT. In addition to physical examination, mid-treatment magnetic resonance imaging or computed tomography (CT) was used to categorize the primary tumor response as complete (CR) or partial (PR).

Results: Ten patients were enrolled. Mean age was 63 years (range: 51-71) and all patients were male. Tumor was located in the tonsil and in the base of tongue in 60% and 40% of patients, respectively. As per AJCC 8th Ed, 50%, 20% and 30% of patients had stage I, II and III, respectively. 40%